

Neurologic Complications of Chronic Kidney Disease

Julian L. Seifter, Li Li, Martin A. Samuels

Disorders of the nervous system may coexist with kidney disease in patients with systemic disorders (e.g., hypertensive encephalopathy, thrombotic microangiopathies, atheroembolic and atherosclerotic disease, vasculitides) and fluid and electrolyte abnormalities. Neurologic complications accompany acute kidney injury (AKI) in the intensive care setting and in outpatients with chronic kidney disease (CKD). A recent meta-analysis revealed that cognitive impairment is evident across the stages of CKD independent of age-related changes.¹ Furthermore, patients with CKD are at increased risk for toxin- and pharmacologic agent-induced neurotoxicity. This chapter focuses on the direct neurologic consequences of CKD.

UREMIC ENCEPHALOPATHY

The syndrome of uremic encephalopathy (UE) involves a spectrum of brain abnormalities that may range from nearly imperceptible changes to coma.

Pathogenesis

The brain in CKD has decreased metabolic activity and oxygen consumption.^{2,3} As long as the underlying kidney disease has not affected cerebral hemodynamics and responsiveness to carbon dioxide, these functions appear intact, but subtle disturbances have been detected after dialysis.

Among the known uremic toxins listed in the European (EUTox) database, 9% (7 of 75 solutes) are associated with neurologic effects.⁴ The balance of excitatory and inhibitory neurotransmitters may be disrupted by organic substances,⁵ in particular guanidino compounds, which are increased in cerebrospinal fluid.^{6,7} These compounds antagonize γ -aminobutyric acid (GABA_A) receptors and at the same time have agonistic effects on *N*-methyl-D-aspartate glutamate receptors, leading to enhanced cortical excitability. Asymmetric dimethylarginine,⁸ which is increased in CKD, inhibits endothelial nitric oxide synthase, and levels correlate with cerebrovascular complications in uremia. The serum level of neuropeptide Y (NPY), a peptide associated with orexigenic (hyperphagic) food intake, was found to be elevated in CKD patients. NPY can diffuse across the blood-brain barrier, and elevated NPY levels in blood promote endothelial dysfunction and regulate neural progenitor cells.^{9,10} Disturbances in monoamine metabolism include a depletion of norepinephrine and suppression of central dopamine, which has been linked to the impairment of motor activity in uremic rats. Myo-inositol, carnitine, indoxyl sulfate, and polyamine content, as well as disrupted solute transport and permeability, have been implicated in the neuronal dysfunction of uremia. Metabolites of drugs such as cimetidine and acyclovir are increased in uremia because of inhibition of organic anion transporter 3 (OAT3) and may result in neurotoxic syndromes.¹¹ Levels of opiates, in particular metabolites of meperidine, are increased in plasma because of decreased

excretion through renal cation secretory transport, with subsequent neurotoxicity.¹²

Secondary hyperparathyroidism also may play a role in UE^{13,14} because brain calcium is increased in CKD, and calcium transporters within neurons are sensitive to parathyroid hormone (PTH). Increased cellular calcium may play a role in neuroexcitation, while PTH may increase alkaline phosphatase levels in the brain, therefore indirectly causing cognitive impairment in CKD patients. High levels of alkaline phosphatase in the brain might promote the binding of tau protein to muscarinic receptors in the hippocampus, leading to a significant increase in cellular calcium level and neuronal cell death.¹⁵ High fibroblast growth factor (FGF)-23, directly through its interaction with α -klotho, was associated with a higher risk of dementia in the Framingham study.¹⁶ On the other hand, low α -klotho levels were associated with increased risk of dementia.¹⁷

Appetite regulation is abnormal in uremia (see [Chapter 90](#)). A high rate of tryptophan entry across the blood-brain barrier may increase the synthesis of serotonin, a major appetite inhibitor.¹⁸ High levels of cholecystokinin, a powerful anorectic, and low levels of NPY, an appetite stimulant, have been observed. Cachexia may result from anorexia, acidosis, and inflammation. Inflammatory cytokines such as leptin, tumor necrosis factor- α , and interleukin-1 may signal anorexic neuropeptides such as proopiomelanocortin and α -melanocyte-stimulating hormone in the arcuate nucleus of the hypothalamus.

Clinical Manifestations

Whereas 20% of critically ill patients with AKI developed neurologic impairment in a study,¹⁹ the syndrome in CKD is more subtle, not correlating closely to the level of kidney function.²⁰ Numerous studies with long-term follow-up in CKD patients suggest that cognitive impairment is greater in patients with albuminuria.^{21,22} Cross-sectional studies in hemodialysis (HD) patients found cognitive impairment in 30%, with about 10% exhibiting severe impairment. Neurocognitive deficits may have special implications for CKD in early childhood, adversely affecting development of the brain.²³

UE can manifest as complex mental changes or motor disturbances ([Box 89.1](#)). The full-blown syndrome is a risk factor for morbidity and mortality.^{2,3} Mental findings include emotional changes, depression, disturbing and disabling cognitive and memory deficits, and, in the most severe form, a generalized disorder characterized by delirium, psychosis, seizures, coma, and ultimately death. Severe motor symptoms or signs are rare. Depression, anxiety, and even suicide are important underdiagnosed and undertreated aspects of uremia and may be related to metabolic or poor nutritional state and fear of dialysis or death. Other known causes of depression should always be sought.

Stable UE manifests with fine action tremor, myoclonus, asterixis, and hyperreflexia. Asterixis is characterized by intermittent loss of

muscle tone in antigravity muscles. It is distinguished from tremor by the fact that it is not an oscillation but rather an intermittent loss of tone. Myoclonus is also seen in patients with UE. It is similar in timing to asterixis (10–100 ms) but is caused by activation of antigravity muscles. For this reason, some consider asterixis to be a form of negative myoclonus. The distinction between asterixis and myoclonus is not as important as once thought, because both or either may be present in many metabolic encephalopathies and some structural brain diseases as well. Asterixis and myoclonus may be elicited with the hands outstretched but detection may be enhanced by looking at the protruded tongue or the index finger raised with the hand resting on a firm surface. Asterixis and myoclonus may be seen in patients with kidney impairment who have received drugs such as metoclopramide, phenothiazines, antiepileptic drugs including gabapentin, and opioids, especially meperidine. It is imperative for the treating physician to understand the life-threatening complications of such drugs for the patient with CKD before and after initiation of dialysis¹² (Table 89.1). Metabolic acidosis may produce an indistinguishable encephalopathy, as can aluminum toxicity. Therefore, a

careful search for other causes is required before features potentially consistent with UE are attributed to advanced uremia requiring kidney replacement therapy (KRT).

Advanced UE is usually characterized by a reduced level of consciousness, anorexia, asterixis, myoclonus, and upper motor neuron signs such as spasticity, dysarthria, and gait disturbance.

Diagnosis and Differential Diagnosis

Severe UE is unlikely to occur in patients with CKD who are followed closely before initiation of KRT (see Chapter 95). UE is more likely to occur in patients who initially present with kidney failure or dialysis patients after missing many treatments. The blood urea nitrogen concentration most commonly exceeds 200 mg/dL. Hypertension is often present, and the manifestation may be mistaken for hypertensive encephalopathy. The diagnosis of UE is based on clinical findings and their improvement after adequate therapy (see next section). If the patient develops more insidious signs of cognitive impairment while undergoing dialysis, inadequate dialysis treatment must be distinguished from dialysis-associated dementia, Alzheimer disease, or other causes of chronic cognitive impairment. Lumbar puncture, electroencephalography, and imaging procedures largely serve to exclude other causes in patients in whom the clinical diagnosis is doubtful. In UE the cerebrospinal fluid is often abnormal, sometimes demonstrating a modest pleocytosis (usually <25 cells/ μ L) and increased protein (usually <100 mg/dL). The electroencephalogram is usually abnormal but nonspecific. Generalized slowing in the theta and delta wave ranges is found.²⁴ Brain imaging often shows cerebral atrophy and enlargement of the ventricles (Fig. 89.1).

The differential diagnosis of UE is shown in Table 89.2. Seizure activity may be secondary to UE, hypertensive encephalopathy, cerebral embolism, cerebral venous thrombosis, or electrolyte and acid-base abnormalities. Tetany can develop when treatment involves alkalization of an acidemic patient with kidney disease and hypocalcemia. Severe electrolyte abnormalities, including hypocalcemia, hypomagnesemia, hyponatremia, and hypophosphatemia, may manifest with delirium.

Treatment

Most nephrologists consider advanced cognitive or memory impairment an indication for initiation of KRT. Most of the manifestations of central nervous system (CNS) involvement are reversible with dialysis within days or weeks, but mild signs of UE may persist. Patients who do not manifest severe signs of encephalopathy often notice improvement in cognitive function once they start KRT. Some may describe this as a “fog lifting.” In dialyzed patients with persistent or recurrent

BOX 89.1 Clinical Manifestations of Uremic Encephalopathy

Early Encephalopathy

Mental Changes

- Mood swings
- Impaired concentration, loss of recent memory
- Insomnia, fatigue, apathy

Motor Changes

- Hyperreflexia
- Tremor, asterixis
- Dysarthria, altered gait, clumsiness, unsteadiness

Late Encephalopathy

Mental Changes

- Altered cognition and perception
- Illusions, visual hallucinations, agitation, delirium
- Stupor, coma

Motor Changes

- Myoclonus, tetany
- Hemiparesis
- Seizures

TABLE 89.1 Opioid Use in End-Stage Kidney Disease

Drug	Active Metabolites	Comments
Codeine	Morphine	Active metabolites accumulate in patients with kidney failure; should avoid in dialysis. Meperidine should not be used in chronic kidney disease because normeperidine causes central nervous system toxicity (seizures, opisthotonos) in kidney disease.
Hydrocodone	Hydromorphone	
Meperidine	Normeperidine	
Morphine	Morphine-3-glucuronide, Morphine-6-glucuronide	Active metabolites accumulate in patients with kidney failure; can cause myoclonus and respiratory depression; mostly replaced by hydromorphone.
Hydromorphone	Hydromorphone-3-glucuronide	Mostly metabolized in liver. Active metabolites removed by dialysis, but small amounts accumulate in patients with kidney failure.
Oxycodone	Oxymorphone	
	Noroxycodone	
Fentanyl	No active metabolites	Metabolized in liver; not removed by dialysis.
Methadone		Fecal excretion; not removed by dialysis.

symptoms, increasing the delivered dialysis dose may improve clinical findings. Successful kidney transplantation usually results in resolution of the UE syndrome within days.

Correction of anemia with recombinant erythropoietin in the dialysis patient (see [Chapter 86](#)) may be associated with improved cognitive function²⁵ and decreased slowing on the electroencephalogram.²⁶ Too rapid overcorrection of anemia may be associated with seizures. Treatment of psychosis in kidney disease must consider the pharmacokinetics of the specific agent. For example, risperidone may be useful,

but dose reduction is necessary because of the prolonged half-life in CKD.

Of note, intensive blood pressure (BP) control may not improve cognitive impairment in CKD patients. A subset of patients ($n = 670$) in the large SPRINT study received magnetic resonance imaging follow-up. Surprisingly, there was an increase in white-matter lesion volume and a decrease in total brain volume in the intensive antihypertensive treatment group compared with the standard treatment group.²⁷

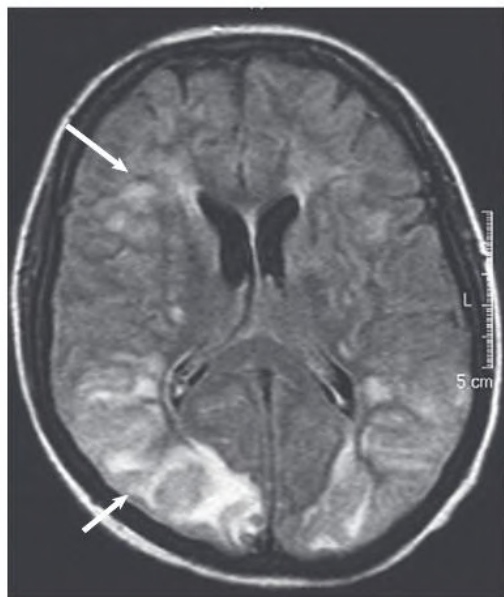


Fig. 89.1 Magnetic Resonance Imaging Findings in Uremic Encephalopathy. Axial T2-weighted (fluid-attenuated inversion recovery) scan from a 40-year-old woman. The extensive hyperintense lesion involves the cortical and subcortical areas of both occipital lobes and, in a more focal distribution, the basal ganglia and the frontal white matter (arrows). The volume of the affected brain parenchyma is increased. Reversibility of the MRI changes was noted 2 weeks after the initiation of regular dialysis. (Courtesy A. Thron, Aachen, Germany.)

PERIPHERAL NEUROPATHY

Patients with CKD are susceptible to both polyneuropathies and mononeuropathies. The pathophysiologic process of polyneuropathy involves axonal degeneration in a length-dependent fashion. Primary demyelinating neuropathies are rare in the context of CKD except when the underlying kidney disease also causes demyelination (e.g., multiple myeloma). Mononeuropathies in CKD may be caused by nerve entrapment with compression of metabolically weakened nerves, particularly in wheelchair users or bed-bound patients. Mononeuritis multiplex should raise the possibility of vasculitic neuropathy, especially when systemic vasculitis (e.g., antineutrophil cytoplasmic antibody–positive small-vessel vasculitis or polyarteritis nodosa) is causing the CKD. Functional sparing of small-diameter axons in uremia is suggested by relatively intact thermal thresholds (hot and cold thermal threshold testing is a surrogate for pain threshold). The modestly slowed nerve conduction velocities in the polyneuropathies of uremia may be related to the reversible inhibition of the sodium-potassium adenosine triphosphatase by a uremic toxin. According to the middle molecule hypothesis, accumulated toxins in the range of 300 to 12,000 Da, including peptide hormones and polyamines, may lead to progression of neuropathy in HD patients.^{6,13,14} Lower limb motor axons in uremic patients are depolarized before but not after dialysis, consistent with a role of hyperkalemia in the development of altered nerve excitability.²⁸ Elevated magnesium levels will also slow nerve conduction velocity. In vitro, extracellular acidosis contributes to decreased sodium conductance in large sensory neurons. Very slow nerve conduction velocities (i.e., less than half normal) suggest a demyelinating

TABLE 89.2 Differential Diagnosis of Uremic Encephalopathy

Differential Diagnosis	Comment
Hemodynamic or Vascular Encephalopathy	
Systemic inflammatory response syndrome	Observed in septic patients.
Systemic vasculitis	Vasculitis or lupus with cerebral involvement.
Cerebral atheroembolic disease	Follows recent aortic or cardiac angiography; associated with peripheral manifestations, including lower extremity cyanosis, livedo reticularis, and eosinophilia.
Drug-Induced Neurotoxicity	
Analgesics	Meperidine, codeine, morphine, gabapentin.
Antibiotics	High-dose penicillins (may cause seizures), acyclovir, ethambutol (optic nerve damage), erythromycin and aminoglycosides (may cause ototoxicity), nitrofurantoin and isoniazid (peripheral neuropathy).
Psychotropics	Lithium, haloperidol, clonazepam, diazepam, chlorpromazine.
Immunosuppressants	Cyclosporine, tacrolimus.
Chemotherapeutics	Cisplatin, ifosfamide.
Others	High doses of loop diuretics (ototoxic), ephedrine, methyl dopa, aluminum metoclopramide (myoclonus, dystonia).
Subdural Hematoma	
Posterior reversible encephalopathy syndrome	Observed particularly after kidney transplantation as a result of reversible, abnormal permeability of the blood-brain barrier. Often manifests with headache followed by mental depression, visual loss, and seizures in the context of volume expansion, acute hypertension, and often treatment with corticosteroids or calcineurin inhibitors. Lesions in the parietal, temporal, and occipital lobes may be seen on imaging studies.

neuropathy, a finding that should lead the physician to seek a specific cause (e.g., a paraprotein).

Uremic neuropathy may progress rapidly in advanced CKD.^{2,28,29} Characteristic symptoms and signs are sensory loss, pain, paresthesias, and insensitivity to temperature, particularly cold. Motor findings such as foot drop can also occur. Phrenic neuropathy may cause dyspnea because of poor diaphragmatic movement, whereas hiccups are more likely a result of the CNS effects of uremia. The distal lower extremities are usually affected first because axonal polyneuropathies are length dependent. Decrease in vibratory sensation and position sense and Romberg sign (i.e., greater instability of stance with eyes closed than with eyes open) are common signs. Muscle stretch reflexes are reduced or absent. In the patient with diabetes who is on dialysis and has progressive neuropathy, it is important to establish adequacy of dialysis, as well as glucose control. Uremic polyneuropathy is aggravated by malnutrition, inadequately controlled hypertension, certain comorbid conditions (e.g., diabetes mellitus, alcohol use, atherosclerotic vascular disease), and medications (e.g., nitrofurantoin, isoniazid, hydralazine).

The diagnosis of uremic polyneuropathy can usually be made from clinical findings. If electromyography and nerve conduction tests are performed, they should not be done in an extremity bearing an arteriovenous fistula because vascular access surgery may cause local nerve injury, which can complicate the interpretation of these studies.²⁸ Nerve conduction velocity is modestly reduced, and needle electromyography shows evidence of chronic denervation and sometimes reinnervation.

Lead polyneuropathy should be considered, particularly when there is a known exposure history. A lower motor neuron syndrome caused by lead toxicity may be mistaken for amyotrophic lateral sclerosis. A bone lead scan using K-line x-ray fluorescence spectroscopy of the tibia is a promising new noninvasive test that may become useful. Serum lead values and red cell protoporphyrin levels may be normal if exposure is remote. There may be associated depression, the so-called *saturnine temperament*, which takes its name from the ancient belief that Saturn was made of lead and was associated with a melancholy disposition. Gout, hypertension, glycosuria, and microcytic anemia also may be caused by lead toxicity.

Other conditions in the differential diagnosis of mixed polyneuropathy include other heavy metals (e.g., arsenic, mercury), nutritional deficiencies (e.g., pyridoxine, folate, thiamine, niacin), HIV-related neuropathy, amyloid, vasculitis, sarcoid, lupus, and a paraneoplastic syndrome.

Progressive polyneuropathy may be an indication for initiation of dialysis or kidney transplantation. Symptoms usually will not deteriorate further or may even show a slow improvement thereafter. If polyneuropathic symptoms worsen in a dialysis patient, the dialysis dose should be increased. Physical therapy is an important component of the management. Patients experiencing neuropathic pain may be treated with tricyclic antidepressants (e.g., amitriptyline 10–25 mg, increasing to 75–150 mg at bedtime) or antiepileptic drugs (e.g., carbamazepine 200–400 mg initially, increasing to 1200 mg maximally; phenytoin 100–200 mg initially, maximally 600 mg).^{2,3} Gabapentin may result in sedation and myoclonus in CKD because of marked prolongation of the drug's half-life, and should be prescribed at reduced doses. Deficiencies of cobalamin (vitamin B₁₂), folate, and pyridoxine may be reflected in an elevated serum homocysteine level. Methylmalonic acid may be elevated in vitamin B₁₂ deficiency, and it is critical to promptly diagnose this deficiency because irreversible changes to the nervous system may develop within months. Loss of balance and position sense may be mistaken for neuropathy in pernicious anemia. Vitamin B₁₂ deficiency and folate deficiency manifest with a megaloblastic or macrocytic anemia. If folate is given to a patient

deficient in vitamin B₁₂, acute posterior and lateral column findings of subacute combined degeneration of the spinal cord may develop as the low store of vitamin B₁₂ is used to correct the anemia. Thiamine deficiency, often associated with malnutrition, can aggravate neuropathy, but whether replacement of any of these vitamins is effective in preventing or curing polyneuropathy in uremic patients is not well established. Thiamine deficiency is the cause of Wernicke encephalopathy in dialysis or malnourished patients. This syndrome is suspected when the triad of mental change (often amnesia), ataxia (most often affecting gait), and oculomotor disturbances (most often abducens palsies with gaze-evoked nystagmus) is seen in any patient whose diet is deficient in B vitamins. When amnesia is combined with a polyneuropathy, the term *Korsakoff syndrome* applies. In malnourished patients, the cause of Korsakoff syndrome is usually multiple subclinical attacks of Wernicke encephalopathy, hence the term *Wernicke-Korsakoff disease*.

Specific mononeuropathic syndromes include ulnar nerve entrapment, associated with uremic tumoral calcinosis and subsequent ischemia, and carpal tunnel syndrome, for example, caused by β_2 -microglobulin–derived amyloidosis (see Chapter 88) or by an arteriovenous fistula.^{2,29} These syndromes may be treated with antiinflammatory agents, antiepileptic drugs, and surgical decompression. It is important to ensure adequacy of dialytic treatment.

Pruritus in the uremic patient may be severe and is not primarily a skin disorder (see also Chapter 91). Rather, it may represent a form of sensory neuropathy. This symptom usually improves with KRT. Antihistamines, with their sedating effect, are not always effective. Gabapentin and carbamazepine block the afferent pathway in uremic neuropathic itch. Gabapentin and pregabalin inhibit calcitonin gene-related peptide from primary afferent neurons by inhibiting GABA and opioid receptor antagonists (naloxone and naltrexone) and may antagonize transmission of itch. Antidepressants also have been successful, possibly by interfering with reuptake of serotonin and norepinephrine to reduce pruritus perception.³⁰

AUTONOMIC NEUROPATHY

Autonomic neuropathy is also very common in patients with advanced CKD, probably because diabetes is a common cause of CKD. Hyperglycemia may be more difficult to control in CKD because glucose filtration is decreased. Amyloidosis, a less common cause of CKD, is associated with autonomic neuropathy. A typical manifestation is orthostatic hypotension, which is most severe in patients with diabetes mellitus or amyloidosis as a cause of CKD. Some dopaminergic drugs used in Parkinson disease or parkinsonism itself may cause orthostasis. Some patients have peripheral neuropathy and may manifest hyporeninemic hypoaldosteronism with hyperkalemia and renal tubular acidosis. Hypotension or orthostasis may preclude treatment with angiotensin antagonists and may complicate fluid removal during dialysis. CKD patients were thought to have decreased baroreceptor function, but normal baroreceptor responses to graded decrements in mean arterial BP have been described.³¹ Instead, CKD patients have sympathetic hyperactivity, which contributes to hypertension, more rapid progression to kidney failure in the predialysis patient, and greater cardiovascular (CV) risk. Accordingly, α - and β -adrenergic blockade has been advocated in CKD.³¹ Autonomic neuropathy is caused by axonal disease and thus is length dependent. For that reason, the longest autonomic nerve, the vagus nerve, is usually the first affected, resulting in the loss of the normal sinus arrhythmia, significant reductions of day-night BP variation, and possibly sudden cardiac death related to the loss of the balance between the sympathetic and parasympathetic limbs of the autonomic nervous system. Reports of gastrointestinal problems include gastroparesis, which is particularly problematic for the diabetic patient. In the

predialysis patient, nausea and early satiety associated with gastroparesis may be confused with uremia. Several medical regimens have been used for the uremic patient with gastroparesis; they include erythromycin, which can activate the gastric motilin receptor. Levodopa-carbidopa, as a dopamine agonist, may be effective, as may metoclopramide 10 mg before sleep or domperidone 10 to 20 mg. Nocturnal diarrhea is another consequence of vagal neuropathy. Erectile dysfunction and incontinence (urinary more commonly than fecal) also may be related to autonomic neuropathy.

CRANIAL NEUROPATHIES

Cranial nerve involvement is most often vestibulocochlear. Hearing loss needs to be distinguished from drug-induced ototoxicity or the neurosensory deafness of hereditary nephropathy.^{2,3} Bilateral vestibular failure leads to inability to stand or to walk normally without vertigo or nystagmus. It is often related to the use of aminoglycoside antibiotics in the patient with CKD. *N*-Acetylcysteine given with aminoglycosides may reduce the risk for cochlear toxicity. Sulfa-based loop diuretics, often used at high doses in patients with CKD, may cause vestibular or cochlear damage. Decreased olfactory function, especially a reduced ability to discriminate among and identify odors, and dysgeusia are commonly seen in patients with CKD.

SLEEP DISORDERS

Sleep disorders in CKD include daytime sleepiness, insomnia, sleep apnea and restless legs syndrome.³² Many HD and peritoneal dialysis patients exhibit obstructive sleep apnea that is independent of obesity.³³ The associated sleep deprivation contributes to fatigue and cognitive impairment and increases the risk for CV complications.³³ Both obstructive and central sleep apnea are prevalent in patients with CKD,³⁴ but the clinical predictive scores used to screen for sleep apnea have poor specificity in CKD patients³²; diagnostic testing with polysomnography and actigraphy systems may be preferable. Obstructive sleep apnea is a condition in which blockage of the upper airway can interfere with nocturnal breathing. Nighttime oxygen and continuous positive airway pressure (CPAP) may help. Treatment of obstructive sleep apnea is effective, including use of CPAP (as in nonuremic individuals)³⁵ and conversion to nocturnal HD.^{36,37} Central sleep apnea, often accompanied by heart failure, causes prolonged apneic episodes³⁸ and may respond to oxygen. Initiation of dialysis may be helpful.

Daytime sleepiness is common and underdiagnosed in patients with CKD and contributes not only to worsened hypertension and increased CV risk but also to social dysfunction. Whether obstructive sleep apnea and its associated excessive daytime sleepiness is an independent risk factor for the progression of kidney failure is not yet settled. It is assessed by a multiple sleep latency test; that is, the duration of time from “lights out” to the onset of sleep. Sleep latency less than 5 minutes is consistent with sleep deprivation. There was also a reduced proportion of rapid eye movement sleep. An increased arousal frequency is related to periodic limb movements during sleep and the presence of sleep apnea.

Sleep-wake complaints are common in patients on dialysis, with an incidence of up to 80%. Contributors include peripheral neuropathy, pain, and pruritus.

RESTLESS LEGS SYNDROME (EKBOM SYNDROME)

Restless legs syndrome (RLS), described by K.A. Ekbom in 1944, is frequent in CKD, particularly in women. It may result from a decrease in dopaminergic modulation of intracortical excitability, with reduced

supraspinal inhibition and increased spinal cord excitability. RLS is characterized by unpleasant “creeping” sensations in the extremities and a compulsive need to move the limbs, usually the legs.^{39,40} The movement is worsened by periods of rest or inactivity and is relieved by walking or stretching. Symptoms are worse at night and may lead to insomnia and consequent daytime sleepiness and reduced quality of life. Nocturnal muscle cramps are common in CKD and should be distinguished from RLS. The Ekbom syndrome consists of restless legs plus other obsessive-compulsive-like disorders including various pica behaviors, such as pagophagia (ice eating), geophagia (clay eating), and amylophagia (starch eating).

Periodic limb movement disorder is characterized by episodes of involuntary repetitive extension of the big toe and dorsiflexion of the ankle, as well as flexion of the knee and hip.⁴⁰ This disorder is more likely to occur in those with RLS.

Iron deficiency or iron transport into the CNS plays a central role in RLS. Iron is a cofactor for the enzyme tyrosine hydroxylase, the rate-limiting step in the biosynthesis of dopamine, possibly explaining the link between iron deficiency and dopamine deficiency in RLS. Overt iron deficiency is easily diagnosed and should be treated.⁴⁰ In patients with normal red blood cell indices and serum iron and total iron-binding capacity, serum ferritin should be tested. Transferrin saturation ratio may be an even more sensitive indicator of iron deficiency. Although not clinically indicated for RLS, analysis of spinal fluid ferritin may reveal a subtle CNS iron deficiency syndrome. RLS often persists after initiation of dialysis but may improve after transplantation and has been linked to abnormalities in calcium and phosphorus metabolism as well as to anemia. Iron replacement should be initiated if there is any indication of iron deficiency. Intravenous iron may be required (see [Chapter 86](#)). Dopaminergic treatment is often helpful, usually starting with the dopamine receptor agonists pramipexole and ropinirole. Levodopa combined with decarboxylase inhibitors (e.g., carbidopa-levodopa) may be used as well as gabapentin, opioids, and benzodiazepines.^{39,40} As mentioned earlier, gabapentin should be used cautiously. Older dopamine receptor agonists, such as bromocriptine and pergolide, are rarely used now for RLS.

NEUROLOGIC SYNDROMES ASSOCIATED WITH KIDNEY REPLACEMENT THERAPY

KRT is associated with an increased incidence of subdural hematoma and intracranial hemorrhage, presumably connected to hypertension and anticoagulation with HD as well as Wernicke encephalopathy (see earlier discussion).^{2,3} A syndrome of muscle weakness has been attributed to L-carnitine depletion from dialysis, leading to decreased mitochondrial fatty acid usage.

Dialysis disequilibrium syndrome is a rare complication of rapid metabolic changes occurring with HD, usually affecting patients in whom HD is being initiated (see [Chapter 100](#)).⁴¹ It is most common in patients with severe uremia of long duration and with severe hypertension. Characterized by acute onset of headache, nausea, vomiting, disorientation, a state of confusion, and seizures, it is a diagnosis of exclusion. It usually results from acute changes in osmolality during HD, in which the rapid decrease in urea in the extracellular fluid favors water movement into brain cells, resulting in cerebral edema. Alternatively, other intracellular osmolytes within brain cells may draw water from the extracellular fluid. The syndrome normally reverses spontaneously after a period of regular HD. If no improvement is seen after a month of HD, one should investigate for other possible causes of the clinical syndrome by imaging the brain, obtaining an electroencephalogram, and examining the spinal fluid. Dialysis disequilibrium syndrome may be prevented by decreasing HD length to 2 to 3 hours, dialyzing daily, providing mannitol during

the dialysis run (such as 25 g IV every 90 minutes) and reducing HD efficacy during the first sessions.

Dialysis encephalopathy (formerly called *dialysis dementia*) is probably a multifactorial syndrome occurring in sporadic-endemic and epidemic types. In particular, in the epidemic type, aluminum-based phosphate binders and exposure to dialysate containing more than 20 µg of aluminum per liter are considered to be major causes.^{42,43} Aluminum transferred to the nervous system by transferrin results in a characteristic clinical condition with prominent stuttering that

usually worsens toward the end of a dialysis session and encephalopathy, initially responding well to IV benzodiazepines but then becoming unresponsive, leading to severe encephalopathy and death. With the almost universal preparation of dialysate water by reverse osmosis and the marked reduction in aluminum-containing phosphate binder use, aluminum-induced encephalopathy has virtually disappeared. If it is present, aluminum toxicity is treated with deferoxamine (see [Chapter 88](#)). Kidney transplantation is an effective treatment for dialysis dementia.

SELF-ASSESSMENT QUESTIONS

1. A 50-year-old man with chronic hypertension, CKD, and eGFR of 8 mL/min/1.73 m² who is not yet on dialysis presents with confusion, lethargy, and myoclonus. Which of the following factors may contribute to these neurologic changes?
 - A. Use of gabapentin for neuropathic pain
 - B. Poorly controlled secondary hyperparathyroidism
 - C. Poorly controlled anemia with iron deficiency
 - D. Poor dietary protein intake resulting in decreased blood urea level
 - E. A and D
 - F. All of the above
2. A 25-year-old woman has hypertension, seizures, and encephalopathy believed to be caused by uremia. Her blood pressure is 200/120 mm Hg. Her blood chemistry reveals Na 120 mEq/L, K 4 mEq/L, Cl 80 mEq/L, and HCO₃⁻ 15 mEq/L. The anion gap is 25 mEq/L, glucose 70 mg/dL, and BUN 200 mg/dL. Which of the following is the *best* answer?
 - A. Hypotonic hyponatremia may contribute to the altered neurologic picture.
 - B. The seizures may be from severe hypertension.
 - C. This presentation is characteristic of uremic encephalopathy.
 - D. The BUN should be decreased to normal promptly with dialysis.
 - E. Both A and D are incorrect.
 - F. A, B, and C are correct.
3. Findings associated with restless legs syndrome include all of the following *except*:
 - A. obsessive-compulsive behavior.
 - B. amylophagia (starch craving).
 - C. iron overload and deposition in the basal ganglia.
 - D. response to dopaminergic medication.
 - E. commonly, improvement during daytime hours and with exercise.

REFERENCES

1. Brodski J, et al. A systematic review of cognitive impairments associated with kidney failure in adults before natural age-related changes. *J Int Neuropsychol Soc.* 2019;25(1):101–114.
2. Brouns R, De Deyn PP. Neurological complications in renal failure: a review. *Clin Neurol Neurosurg.* 2004;107(1):1–16.
3. Burn DJ, Bates D. Neurology and the kidney. *J Neurol Neurosurg Psychiatry.* 1998;65(6):810–821.
4. Viggiano D, et al. Mechanisms of cognitive dysfunction in CKD. *Nat Rev Nephrol.* 2020;16(8):452–469.
5. Smogorzewski MJ. Central nervous dysfunction in uremia. *Am J Kidney Dis.* 2001;38(4 suppl 1):S122–S128.
6. Vanholder R, et al. New insights in uremic toxins. *Kidney Int Suppl.* 2003;(84):S6–S10.
7. De Deyn PP, et al. Endogenous guanidino compounds as uremic neurotoxins. *Kidney Int Suppl.* 2001;78:S77–S83.
8. De Deyn PP, Vanholder R, D'Hooge R. Nitric oxide in uremia: effects of several potentially toxic guanidino compounds. *Kidney Int Suppl.* 2003;(84):S25–S8.
9. Choi B, et al. Elevated neuropeptide Y in endothelial dysfunction promotes macrophage infiltration and smooth muscle foam cell formation. *Front Immunol.* 2019;10:1701.
10. Geloso MC, et al. Cellular targets for neuropeptide Y-mediated control of adult neurogenesis. *Front Cell Neurosci.* 2015;9:85.
11. Ohtsuki S, et al. Role of blood-brain barrier organic anion transporter 3 (OAT3) in the efflux of indoxyl sulfate, a uremic toxin: its involvement in neurotransmitter metabolite clearance from the brain. *J Neurochem.* 2002;83(1):57–66.
12. Smith HS. Opioid metabolism. *Mayo Clin Proc.* 2009;84(7):613–624.
13. Fraser CL, Arief AI. Nervous system complications in uremia. *Ann Intern Med.* 1988;109(2):143–153.
14. Lockwood AH. Neurologic complications of renal disease. *Neurol Clin.* 1989;7(3):617–627.
15. Díaz-Hernández M, et al. Tissue-nonspecific alkaline phosphatase promotes the neurotoxicity effect of extracellular tau. *J Biol Chem.* 2010;285(42):32539–32548.
16. McGrath ER, et al. Circulating fibroblast growth factor 23 levels and incident dementia: the Framingham heart study. *PLoS One.* 2019;14(3):e0213321.
17. Kuriyama N, et al. Association between α -klotho and deep white matter lesions in the brain: a pilot case control study using brain MRI. *J Alzheimers Dis.* 2018;61(1):145–155.
18. Aguilera A, et al. Eating behavior disorders in uremia: a question of balance in appetite regulation. *Semin Dial.* 2004;17(1):44–52.
19. Mehta RL, et al. Spectrum of acute renal failure in the intensive care unit: the PICARD experience. *Kidney Int.* 2004;66(4):1613–1621.
20. Tyler HR. Neurologic disorders seen in the uremic patient. *Arch Intern Med.* 1970;126(5):781–786.
21. Ekblad LL, et al. Albuminuria and microalbuminuria as predictors of cognitive performance in a general population: an 11-year follow-up study. *J Alzheimers Dis.* 2018;62(2):635–648.
22. Martens RJ, Estimated GFR, et al. Albuminuria, and cognitive performance: the maastricht study. *Am J Kidney Dis.* 2017;69(2):179–191.
23. Gipson DS, et al. The nervous system and chronic kidney disease in children. *Pediatr Nephrol.* 2004;19(8):832–839.
24. Balzar E, et al. Quantitative EEG: investigation in children with end stage renal disease before and after haemodialysis. *Clin Electroencephalogr.* 1986;17(4):195–202.
25. Vinothkumar G, et al. Correlation between abnormal GSK3 β , β amyloid, total tau, p-tau 181 levels and neuropsychological assessment total scores in CKD patients with cognitive dysfunction: impact of rHuEPO therapy. *J Clin Neurosci.* 2019;69:38–42.
26. Stivelman JC. Benefits of anaemia treatment on cognitive function. *Nephrol Dial Transplant.* 2000;15(suppl 3):29–35.
27. Nasrallah IM, et al. Association of intensive vs standard blood pressure control with cerebral white matter lesions. *JAMA.* 2019;322(6):524–534.
28. Krishnan AV, et al. Altered motor nerve excitability in end-stage kidney disease. *Brain.* 2005;128(Pt 9):2164–2174.
29. Krishnan AV, et al. Sensory nerve excitability and neuropathy in end stage kidney disease. *J Neurol Neurosurg Psychiatry.* 2006;77(4):548–551.
30. Patel TS, Freedman BI, Yosipovitch G. An update on pruritus associated with CKD. *Am J Kidney Dis.* 2007;50(1):11–20.
31. Koomans HA, Blankestijn PJ, Joles JA. Sympathetic hyperactivity in chronic renal failure: a wake-up call. *J Am Soc Nephrol.* 2004;15(3):524–537.
32. Maung SC, et al. Sleep disorders and chronic kidney disease. *World J Nephrol.* 2016;5(3):224–232.
33. Hanly P. Sleep apnea and daytime sleepiness in end-stage renal disease. *Semin Dial.* 2004;17(2):109–114.
34. Lin CH, Lurie RC, Lyons OD. Sleep apnea and chronic kidney disease: a state-of-the-art review. *Chest.* 2020;157(3):673–685.
35. Marrone O, et al. Fixed but not autoadjusting positive airway pressure attenuates the time-dependent decline in glomerular filtration rate in patients with OSA. *Chest.* 2018;154(2):326–334.
36. Hanly PJ, Pierratos A. Improvement of sleep apnea in patients with chronic renal failure who undergo nocturnal hemodialysis. *N Engl J Med.* 2001;344(2):102–107.
37. Tang SC, et al. Alleviation of sleep apnea in patients with chronic renal failure by nocturnal cyclical-assisted peritoneal dialysis compared with conventional continuous ambulatory peritoneal dialysis. *J Am Soc Nephrol.* 2006;17(9):2607–2616.
38. Yumino D, et al. Nocturnal rostral fluid shift: a unifying concept for the pathogenesis of obstructive and central sleep apnea in men with heart failure. *Circulation.* 2010;121(14):1598–1605.
39. Earley CJ. Clinical practice. Restless legs syndrome. *N Engl J Med.* 2003;348(21):2103–2109.
40. Barrière G, et al. The restless legs syndrome. *Prog Neurobiol.* 2005;77(3):139–165.
41. Arief AI. Dialysis disequilibrium syndrome: current concepts on pathogenesis and prevention. *Kidney Int.* 1994;45(3):629–635.
42. Alfrey AC. Dialysis encephalopathy. *Kidney Int Suppl.* 1986;18: S53–7.
43. Nayak P. Aluminum: impacts and disease. *Environ Res.* 2002;89(2):101–115.